

 Item Navigation

Introduction to Image Stitching

Read the text below. Then navigate to the "Module 4" folder to open **IntroductionToImageStitching.mlx**, included with the course files, to stitch together two images.

Using image registration to align images is important when comparing the same scene or combining multiple images of the same scene. Examples include aligning satellite images taken at different times and aligning different types of medical images.

But what if the goal is to put images of different scenes together? The images must still have some overlap so that you can identify control points to register the images, but there are additional steps. The image stitching steps are:

1. Register pairs of images. The main output you need from the registration is the geometric transformation from each image to your chosen fixed image.
2. Initialize a template for the larger panoramic image. You do this using the geometric transformations to determine the size of the final panorama and the new pixel locations for each image in the panorama.
3. Create the panorama by warping the images and placing them into the correct locations in the new panoramic image.

Step 2 requires converting the pixel locations from rows and columns in a matrix to locations in a "world coordinate" system. The "world coordinate" system describes how to place the pixels in the warped image relative to the fixed image.

In step 3, you warp the images and specify how to place them into the larger panoramic image, including what to do with pixels in the overlapping regions.

If that sounds confusing to you, don't worry. MATLAB includes several functions to help you achieve this task. Open the **IntroductionToImageStitching.mlx** and work through the script to learn how to use these functions. Then, watch the video lesson to learn how to stitch many images together.